RAGHAVENDRA CHANDRAKANTHA BHAT

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PROFESSIONAL SUMMARY

Motivated MCA student with solid software engineering fundamentals in **Python, Java, and SQL**, alongside hands-on experience developing production-grade **Retrieval-Augmented Generation (RAG)** systems and end-to-end Machine Learning pipelines. Proven ability in building offline local LLM workflows, FAISS vector similarity search, and automated data extraction pipelines. Seeking an entry-level **Software Developer / SDE, AI/ML Engineer, or Backend Developer** role.

TECHNICAL SKILLS

Languages:	Python, Java, SQL, C, HTML5, CSS3, JavaScript
AI & GenAI:	Retrieval-Augmented Generation (RAG), LangChain, FAISS Vector Search, Ollama, Vector Embeddings
ML & Data:	Scikit-learn, Pandas, NumPy, Data Preprocessing, Model Evaluation, Random Forest
Databases & Tools:	MySQL, Streamlit, Git/GitHub, PyMuPDF, Tesseract OCR, VS Code, Eclipse

EDUCATION

Master of Computer Applications (MCA) | Bangalore University, Bengaluru *2024 – 2026 (Expected)*

Bachelor of Computer Applications (BCA) | Vasavi Jnana Peetha Degree College, Bengaluru *2021 – 2024* | ***CGPA: 8.91***

PROJECTS

uniRAG — University Academic RAG & Exam Intelligence System *Python, LangChain, FAISS, Ollama, Streamlit*

- Built a 100% offline, privacy-first RAG pipeline using 4-bit quantized Mistral-7B and Nomic dense embeddings indexing 22 MCA subjects.
- Implemented a 4-level hybrid search engine (Exact, Subtopic, Vector Match, Filters) over 1,420+ past exam questions with trend analytics.
- Engineered an intelligent query understanding parser with deterministic syllabus-grounding guards that eliminate hallucinations in 0.00s.
- Developed multi-tier SHA-256 caching and marks-calibrated token budgeting (2–10 marks), slashing cached query response times to sub-second latency.
- Deployed an interactive Streamlit UI supporting real-time SSE token streaming, question paper PDF solvers, and role-based admin security.

Student Performance Prediction System *Python, Scikit-learn, Pandas, Jupyter*

- Trained a Random Forest classification model achieving **86.5% accuracy** in predicting student semester outcomes and academic success.
- Preprocessed and analyzed 500+ student records across 10+ indicators including SGPA, backlogs, attendance, and internal marks.
- Evaluated model performance using confusion matrices, precision, recall, and k-fold cross-validation to prevent feature overfitting.

Flight Delay Prediction System *Python, Scikit-learn, Machine Learning, Pickle*

- Developed a predictive machine learning pipeline to forecast commercial flight departure delays based on weather data and carrier schedules.
- Performed exploratory data analysis (EDA), handled missing values, categorical encoding, and serialized trained models using Pickle.

CERTIFICATIONS

AI Foundations Associate – Oracle *(Generative AI, OCI AI Services, Machine Learning, Deep Learning)* *Sep 2025*

Web Developer – NSDC / Skill India *(OOP Concepts, Java Application Development, Web Fundamentals)* *Nov 2024*